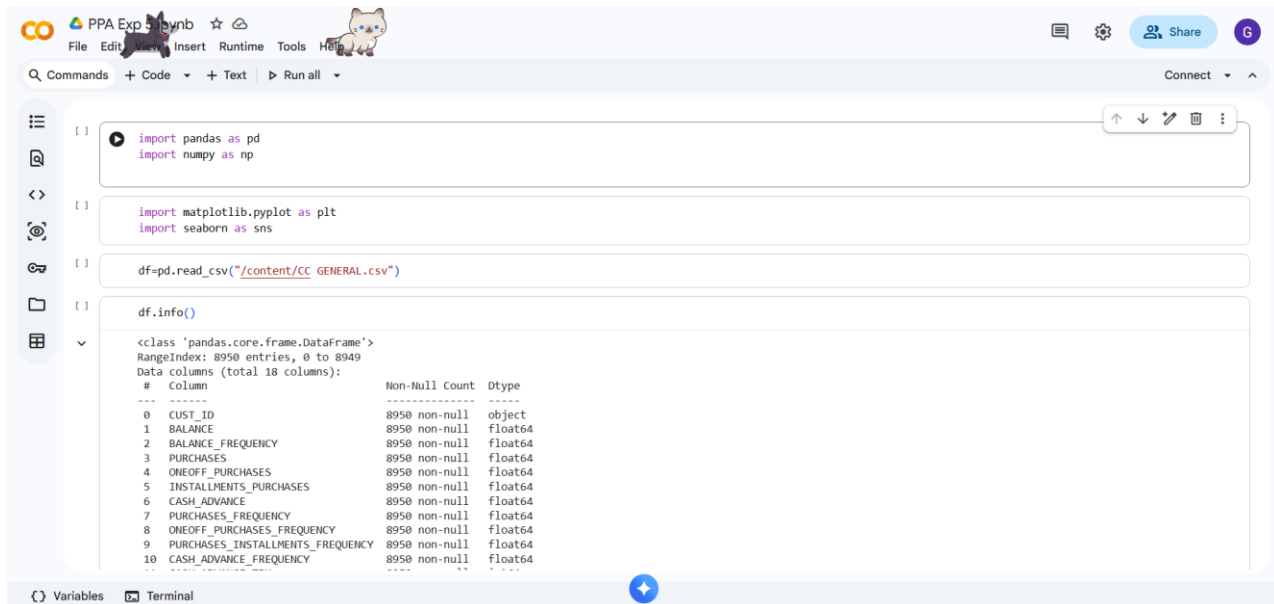


Clustering based data analytics using R/Python. (K-Means, SOM algorithms)

Program:



The screenshot shows a Jupyter Notebook interface with the following code cells:

```
[ ] import pandas as pd
import numpy as np

[ ] import matplotlib.pyplot as plt
import seaborn as sns

[ ] df=pd.read_csv("/content/CC_GENERAL.csv")

[ ] df.info()
```

The output of the `df.info()` cell is displayed below the code:

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 8950 entries, 0 to 8949
Data columns (total 18 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   CUST_ID                               8950 non-null   object
1   BALANCE                               8950 non-null   float64
2   BALANCE_FREQUENCY                     8950 non-null   float64
3   PURCHASES                             8950 non-null   float64
4   ONEOFF_PURCHASES                      8950 non-null   float64
5   INSTALLMENTS_PURCHASES                8950 non-null   float64
6   CASH_ADVANCE                          8950 non-null   float64
7   PURCHASES_FREQUENCY                   8950 non-null   float64
8   ONEOFF_PURCHASES_FREQUENCY            8950 non-null   float64
9   PURCHASES_INSTALLMENTS_FREQUENCY      8950 non-null   float64
10  CASH_ADVANCE_FREQUENCY                 8950 non-null   float64
```

```
import pandas as pd import numpy as np import
matplotlib.pyplot as plt import seaborn as sns
from sklearn.preprocessing import StandardScaler
from sklearn.cluster import KMeans from
sklearn.metrics import silhouette_score df =
pd.read_csv("/content/CC_GENERAL.csv") df.head()
df.isnull().sum()
```

...	CUST_ID	BALANCE	BALANCE_FREQUENCY	PURCHASES	ONEOFF_PURCHASES	INSTALLMENTS_PURCHASES	CASH_ADVANCE
0	C10001	40.900749	0.818182	95.40	0.00	95.4	0.000000
1	C10002	3202.467416	0.909091	0.00	0.00	0.0	6442.945483
2	C10003	2495.148862	1.000000	773.17	773.17	0.0	0.000000
3	C10004	1666.670542	0.636364	1499.00	1499.00	0.0	205.788017
4	C10005	817.714335	1.000000	16.00	16.00	0.0	0.000000

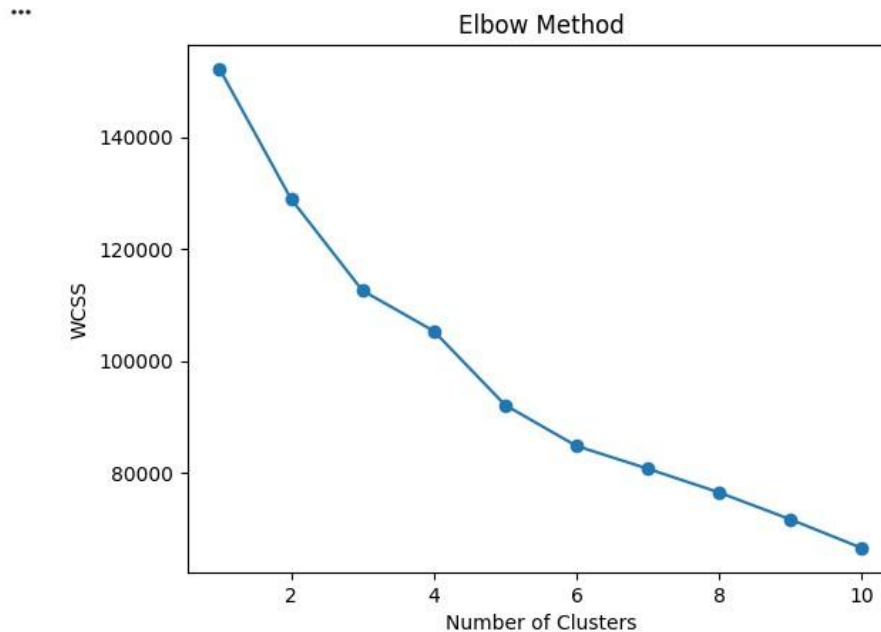
Next steps: [Generate code with df](#) [New interactive sheet](#)

 df.isnull().sum()

...	0
CUST_ID	0
BALANCE	0
BALANCE_FREQUENCY	0
PURCHASES	0
ONEOFF_PURCHASES	0
INSTALLMENTS_PURCHASES	0
CASH_ADVANCE	0
PURCHASES_FREQUENCY	0
ONEOFF_PURCHASES_FREQUENCY	0
PURCHASES_INSTALLMENTS_FREQUENCY	0
CASH_ADVANCE_FREQUENCY	0
CASH_ADVANCE_TRX	0
PURCHASES_TRX	0
CREDIT_LIMIT	1
PAYMENTS	0
MINIMUM_PAYMENTS	313
PRC_FULL_PAYMENT	0
TENURE	0

```
df.fillna(df.mean(), inplace=True)
df.drop('CUST_ID', axis=1, inplace=True)
df.fillna(df.mean(), inplace=True)
scaler = StandardScaler() scaled_data =
scaler.fit_transform(df) wcss = [] for
i in range(1, 11):
    kmeans = KMeans(n_clusters=i, random_state=42)
    kmeans.fit(scaled_data)
wcss.append(kmeans.inertia_)
```

```
plt.plot(range(1, 11), wcss, marker='o')
plt.xlabel("Number of Clusters")
plt.ylabel("WCSS") plt.title("Elbow
Method") plt.show()
```



```
for i in range(2, 11):
    kmeans = KMeans(n_clusters=i, random_state=42)
    labels = kmeans.fit_predict(scaled_data)
    print("Clusters:", i, "Silhouette Score:",
    silhouette_score(scaled_data, labels))
```

```
Clusters: 2 Silhouette Score: 0.27991297227096823
Clusters: 3 Silhouette Score: 0.20676101192444302
Clusters: 4 Silhouette Score: 0.16639208656701637
Clusters: 5 Silhouette Score: 0.19234566924305485
Clusters: 6 Silhouette Score: 0.202755081788085
Clusters: 7 Silhouette Score: 0.19486979124151635
Clusters: 8 Silhouette Score: 0.1899728747613868
Clusters: 9 Silhouette Score: 0.2016552706516606
Clusters: 10 Silhouette Score: 0.2205181120683996
kmeans = KMeans(n_clusters=4, random_state=42)
clusters = kmeans.fit_predict(scaled_data)
df['Cluster'] = clusters
cluster_profile =
df.groupby('Cluster').mean() cluster_profile
```

	BALANCE	BALANCE_FREQUENCY	PURCHASES	ONEOFF_PURCHASES	INSTALLMENTS_PURCHASES	CASH_ADVANCE
Cluster						
0	5023.958508	0.967107	721.184928	453.485000	267.804742	5075.737329
1	109.464707	0.412883	351.589913	191.487250	160.571811	299.684391
2	1253.422099	0.974766	2181.418772	1241.370362	940.311888	234.489319
3	1500.348531	0.966640	282.216074	209.226223	73.305481	789.088912

Silhouette Score

```
for i in range(2, 11):
    kmeans = KMeans(n_clusters=i, random_state=42)
    labels = kmeans.fit_predict(scaled_data)
    print("Clusters:", i, "Silhouette Score:",
          silhouette_score(scaled_data, labels))
Clusters: 2 Silhouette Score: 0.27991297227096823
Clusters: 3 Silhouette Score: 0.20676101192444302
Clusters: 4 Silhouette Score: 0.16639208656701637
Clusters: 5 Silhouette Score: 0.19234566924305485
Clusters: 6 Silhouette Score: 0.202755081788085
Clusters: 7 Silhouette Score: 0.19486979124151635
Clusters: 8 Silhouette Score: 0.1899728747613868
Clusters: 9 Silhouette Score: 0.2016552706516606
Clusters: 10 Silhouette Score: 0.2205181120683996
```

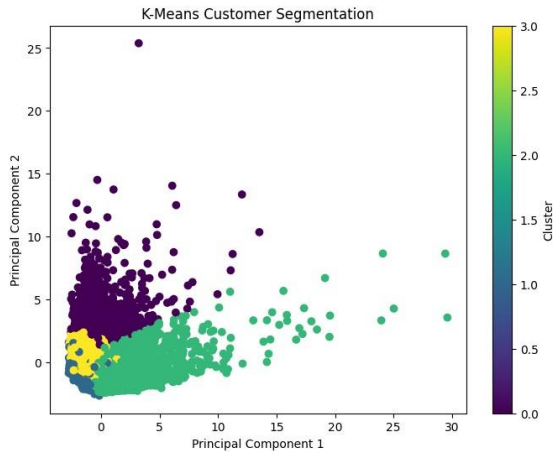
```
cluster_profile = df.groupby('Cluster').mean()
cluster_profile
```

	BALANCE	BALANCE_FREQUENCY	PURCHASES	ONEOFF_PURCHASES	INSTALLMENTS_PURCHASES	CASH_ADVANCE
Cluster						
0	5023.958508	0.967107	721.184928	453.485000	267.804742	5075.737329
1	109.464707	0.412883	351.589913	191.487250	160.571811	299.684391
2	1253.422099	0.974766	2181.418772	1241.370362	940.311888	234.489319
3	1500.348531	0.966640	282.216074	209.226223	73.305481	789.088912

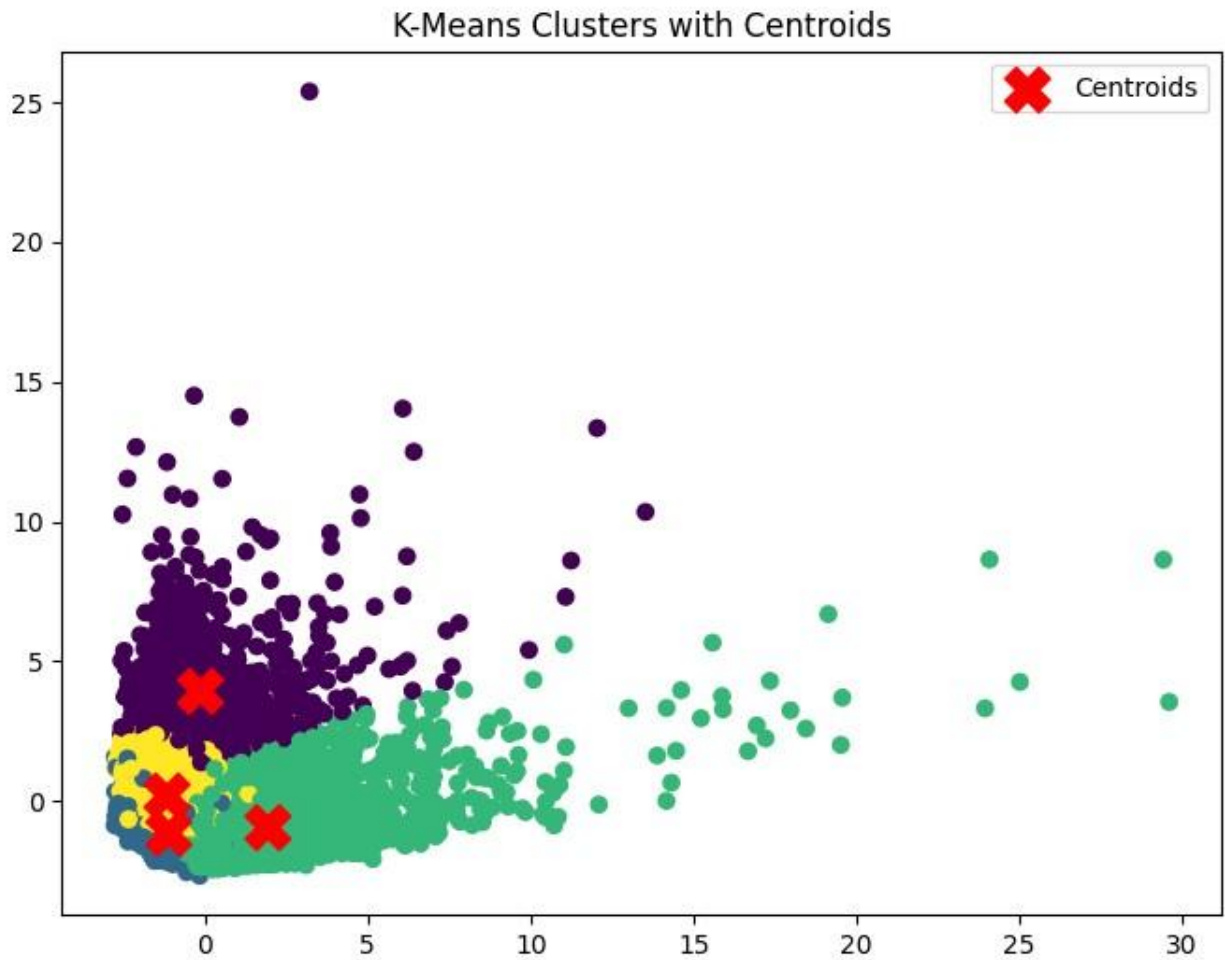
For Visualization

```
from sklearn.decomposition import PCA  pca = PCA(n_components=2)
pca_data = pca.fit_transform(scaled_data)  import matplotlib.pyplot
as plt import seaborn as sns  plt.figure(figsize=(8,6))
```

```
plt.scatter(pca_data[:,0], pca_data[:,1], c=clusters, cmap='viridis')
plt.title("K-Means Customer Segmentation") plt.xlabel("Principal
Component 1") plt.ylabel("Principal Component 2")
plt.colorbar(label="Cluster") plt.show()
```



```
kmeans = KMeans(n_clusters=4, random_state=42) clusters =
kmeans.fit_predict(scaled_data) centroids = kmeans.cluster_centers_
centroids_pca = pca.transform(centroids) plt.figure(figsize=(8,6))
plt.scatter(pca_data[:,0], pca_data[:,1], c=clusters, cmap='viridis')
plt.scatter(centroids_pca[:,0], centroids_pca[:,1],
s=300, c='red', marker='X', label='Centroids') plt.title("K-Means
Clusters with Centroids") plt.legend() plt.show()
```



```
plt.figure(figsize=(10,6)) sns.heatmap(cluster_profile,  
annot=True, cmap='coolwarm') plt.title("Cluster Behavior  
Heatmap") plt.show()
```

